

Homework Problems

Aniruddha Madhava

Fall 2026

Contents

1	Overview	2
2	Functional and Harmonic Analysis	3
2.1	Stein & Shakarchi	3
	Problem 1.1	3
	Problem 1.8	4
	Problem 1.12	5
	Problem 2.2	6
	Problem 2.3	7
	Problem 2.10	8
	Problem 2.16	9
	Problem 2.20	10
2.2	Evans	11
	Problem 2.18	11
	Problem 2.19	11
	Problem 2.21	12
2.3	Luk Notes	14
	Problem N.1	14
	Problem N.2	14
	Problem N.3	15
	Problem E.1.5	15
	Problem E.1.6	16
	Problem E.1.9	18
	Problem E.1.12	18
2.4	Random Computations	19
	Problem 2	19

⁰Last updated: 2026-10-06 at 15:04

1 Overview

The following document is a compilation of various calculations, computations, and exercises I completed as part of orals prep. The work here is not meant to be an extensive account of my work, by any means.

2 Functional and Harmonic Analysis

2.1 Stein & Shakarchi

Problem 1.1. Consider $L^p = L^p(\mathbb{R}^d)$ with Lebesgue measure. Let $f_0(x) = |x|^{-\alpha}$ if $|x| < 1$, $f_0(x) = 0$ for $|x| \geq 1$; also let $f_\infty(x) = |x|^{-\alpha}$ if $|x| \geq 1$, $f_\infty(x) = 0$ when $|x| < 1$. Show that:

- (a) $f_0 \in L^p$ if and only if $p\alpha < d$.
- (b) $f_\infty \in L^p$ if and only if $d < p\alpha$.
- (c) What happens if in the definitions of f_0 and f_∞ we replace $|x|^{-\alpha}$ by $|x|^{-\alpha}/(\log(2/|x|))$ for $|x| < 1$, and $|x|^{-\alpha}$ by $|x|^{-\alpha}/(\log(2|x|))$ for $|x| \geq 1$?

- (a) Let $p \geq 1$, and define $f_0(x) : \mathbb{R}^d \rightarrow \mathbb{R}$ to be the measurable function

$$f_0(x) = \begin{cases} |x|^{-\alpha}, & |x| < 1 \\ 0, & |x| \geq 1. \end{cases}$$

Then $f_0(x) \in L^p$ if and only if $\int_{|x| \leq 1} |f_0(x)|^p dx = \int_{|x| \leq 1} |x|^{-p\alpha} dx < \infty$. To evaluate this integral, we will use integration in polar coordinates. Using Theorem 2.49 from Folland, one has

$$\int_{|x| \leq 1} \frac{1}{|x|^{p\alpha}} dx = \Gamma(S^{d-1}) \cdot \int_0^1 r^{-p\alpha+d-1} dr.$$

By calculus in one dimension, we know that the integral on the right side is finite if and only if $-p\alpha + d - 1 > -1$, which implies $p\alpha < d$. Thus, $f_0 \in L^p$ if and only if $p\alpha < d$.

- (b) Define $f_\infty(x) : \mathbb{R}^d \rightarrow \mathbb{R}$ to be measurable function

$$f_\infty(x) = \begin{cases} 0, & |x| < 1 \\ |x|^{-\alpha}, & |x| \geq 1. \end{cases}$$

Then $f_\infty(x) \in L^p$ if and only if $\int_{|x| \geq 1} |f_\infty(x)|^p dx = \int_{|x| \geq 1} |x|^{-p\alpha} dx < \infty$. Using Theorem 2.49 from Folland, one has

$$\int_{|x| \geq 1} \frac{1}{|x|^{p\alpha}} dx = \Gamma(S^{d-1}) \cdot \int_1^\infty r^{-p\alpha+d-1} dr.$$

By calculus in one dimension, we know that the integral on the right side is finite if and only if $-p\alpha + d - 1 < -1$, which implies $p\alpha > d$. Thus, $f_\infty \in L^p$ if and only if $p\alpha > d$.

- (c) Now define $f_0(x) : \mathbb{R}^d \rightarrow \mathbb{R}$ to be the modified function

$$f_0(x) = \begin{cases} \frac{|x|^{-\alpha}}{\log(2/|x|)}, & |x| < 1 \\ 0, & |x| \geq 1. \end{cases}$$

As in (a), $f_0(x) \in L^p$ if and only if

$$\int_0^1 \frac{r^{-p\alpha+d-1}}{\log(2/r)^p} dr < \infty.$$

Let $k := -p\alpha + d - 1$, and define $u = \log(2/r)$. Then the above integral transforms into

$$\int_0^1 \frac{r^{-p\alpha+d-1}}{\log(2/r)^p} dr = - \int_{\log(2)}^\infty \frac{2^k e^{-ku}}{u^p} (-2e^{-u} du) = 2^{k+1} \int_{\log(2)}^\infty \frac{e^{-(k+1)u}}{u^p} du.$$

In the limit $u \rightarrow \infty$, if $p > 1$, the integrand decays to zero provided $k + 1 \geq 0$. Thus, $f_0(x) \in L^p$ if and only if $k \geq -1$, which is equivalent to requiring $p\alpha \leq d$. If $p = 1$, then $f_0(x) \in L^p$ if and only if $p\alpha < d$. Now define the modified function $f_\infty(x) : \mathbb{R}^d \rightarrow \mathbb{R}$ by

$$f_\infty(x) = \begin{cases} 0, & |x| < 1 \\ \frac{|x|^{-\alpha}}{\log(2|x|)}, & |x| \geq 1. \end{cases}$$

As in (b), $f_\infty(x) \in L^p$ if and only if the integral

$$\int_1^\infty \frac{r^{-p\alpha+d-1}}{\log(2r)} dr < \infty.$$

Taking $u = \log(2r)$ this time, the above integral transforms into

$$\int_1^\infty \frac{r^{-p\alpha+d-1}}{\log(2r)^p} dr = \int_{\log(2)}^\infty \frac{2^{-k} e^{ku}}{u^p} (2^{-1} e^u du) = 2^{-(k+1)} \int_{\log(2)}^\infty \frac{e^{(k+1)u}}{u^p} du.$$

Here, we must consider two cases. First, suppose $p > 1$. Then the integrand decays to zero provided $k+1 \leq 0$ (corresponding to $p\alpha \geq d$). Thus $f_\infty(x) \in L^p$ for $p > 1$ if and only if $p\alpha \geq d$. On the other hand, if $p = 1$, then the integral is finite if and only if $k+1 \not\leq 0$, which corresponds to $p\alpha < d$. Thus,

$$f_\infty(x) \in L^p(\mathbb{R}^d) \text{ if and only if } \begin{cases} p\alpha \geq d, & p > 1 \\ \alpha > d, & p = 1. \end{cases}$$

Problem 1.8. Suppose $1 \leq p < \infty$, and that $\mathbb{R}^d$ is equipped with Lebesgue measure. Show that if $f \in L^p(\mathbb{R}^d)$, then

$$\|f(x+h) - f(x)\|_{L^p} \rightarrow 0 \quad \text{as } |h| \rightarrow 0.$$

Prove that this fails when $p = \infty$.

[Hint: By the previous exercise, the continuous functions with compact support are dense in $L^p(\mathbb{R}^d)$ for $1 \leq p < \infty$.]

Fix $1 \leq p < \infty$, and assume that $\mathbb{R}^d$ is equipped with Lebesgue measure. Further fix some $f \in L^p(\mathbb{R}^d)$. By the density of $C_c(\mathbb{R}^d)$ in $L^p(\mathbb{R}^d)$, there exists a sequence $\{g_j\}_{j=1}^\infty$ of compactly supported continuous functions so that $g_j \rightarrow f$ under the L^p -norm. First, we will show that for every j , $\|g_j(h+\cdot) - g_j\|_p \rightarrow 0$ as $h \rightarrow 0$. Without loss of generality, let $|h| \leq R$ for some $R > 0$. Then by continuity of g_j , $|g_j(x+h) - g_j(x)|^p \rightarrow 0$ pointwise a.e. as $|h| \rightarrow 0$. On the other hand, by continuity of g_j and the extreme value theorem, one has

$$|g_j(x+h) - g_j(x)|^p \leq 2^{p-1}(|g_j(x+h)|^p + |g_j(x)|^p) \leq M \chi_{K+B_R(0)} \in L^1(\mathbb{R}^d),$$

where $M = 2^p \cdot (\max_K |g_j(x)|)^p$. Thus, by the Dominated Convergence Theorem,

$$\int_{\mathbb{R}^d} |g_j(x+h) - g_j(x)|^p dx \xrightarrow{|h| \rightarrow 0} 0.$$

Now fix $\varepsilon > 0$, j large enough so that $\|f - g_j\|_p^p < \varepsilon/3$, and let $\delta > 0$ so that for all $|h| < \delta$, one has $\|g_j(h+\cdot) - g_j\|_p^p < \varepsilon/3$. Then we observe that

$$\begin{aligned} \|f(h+\cdot) - f\|_p^p &= \int_{\mathbb{R}^d} |f(x+h) - f(x)|^p dx \\ &\leq C \cdot \left[\int_{\mathbb{R}^d} |f(x+h) - g_j(x+h)|^p dx + \int_{\mathbb{R}^d} |g_j(x+h) - g_j(x)|^p dx + \int_{\mathbb{R}^d} |g_j(x) - f(x)|^p dx \right] \\ &= 2C \|f - g_j\|_p^p + C \cdot \|g_j(h+\cdot) - g_j\|_p^p \\ &< C\varepsilon. \end{aligned}$$

where in going from the second to the third line, we used translation invariance of the L^p norms. Since the left side is independent of ε , taking the limit $\varepsilon \rightarrow 0$ shows that $\|f(h+\cdot) - f\|_p^p \rightarrow 0$ as $|h| \rightarrow 0$.

Now consider the case of $p = \infty$. Let $f \in L^\infty(\mathbb{R}^d)$ be the following function:

$$f(x) = \begin{cases} 1, & |x| \geq 1 \\ -1, & |x| < 1. \end{cases}$$

Then for any $|h| > 0$, one has that $\|f(h+\cdot) - f\|_{L^\infty} = 2$ so that this norm cannot converge to zero.

Problem 1.12. Suppose the measure space (X, μ) is separable as defined in Exercise 10. Let $1 \leq p < \infty$ and $1/p + 1/q = 1$. A sequence $\{f_n\}$ with $f_n \in L^p$ is said to converge to $f \in L^p$ **weakly** if

$$\int f_n g \, d\mu \rightarrow \int f g \, d\mu \quad \text{for every } g \in L^q.$$

- (a) Verify that if $\|f - f_n\|_{L^p} \rightarrow 0$, then f_n converges to f weakly.
- (b) Suppose $\sup_n \|f_n\|_{L^p} < \infty$. Then, to verify weak convergence, it suffices to check the above convergence for a dense subset of functions g in L^q .
- (c) Suppose $1 < p < \infty$. Show that if $\sup_n \|f_n\|_{L^p} < \infty$, then there exists $f \in L^p$ and a subsequence $\{n_k\}$ so that f_{n_k} converges weakly to f .

- (a) Assume that $\|f - f_n\|_{L^p} \rightarrow 0$. Fix an arbitrary $g \in L^q$. Then by Hölder's inequality,

$$\left| \int f_n g \, d\mu - \int f g \, d\mu \right| \leq \int |f_n - f| |g| \, d\mu \leq \|f_n - f\|_p \|g\|_q \xrightarrow{n \rightarrow \infty} 0.$$

Since $g \in L^q$ was arbitrary, we conclude that f_n converges to f weakly.

- (b) Suppose $\sup_n \|f_n\|_{L^p} < \infty$, and further assume that the above result holds for some dense subset $\mathcal{A}$ in L^q . Fix an arbitrary $g \in L^q$, and let $\{g_j\}_{j=1}^\infty \subset \mathcal{A}$ so that $\|g - g_j\|_q \rightarrow 0$ as $j \rightarrow \infty$. Then

$$\begin{aligned} \left| \int f_n g - \int f g \right| &= \left| \int f_n g - \int f_n g_j + \int f_n g_j - \int f g_j + \int f g_j - \int f g \right| \\ &\leq \int |f_n| |g - g_j| + \left| \int f_n g_j - \int f g_j \right| + \int |f| |g - g_j| \\ &\leq \|f_n\|_{L^p} \|g - g_j\|_{L^q} + \left| \int f_n g_j - \int f g_j \right| + \|f\|_{L^p} \|g - g_j\|_{L^q} \\ &\leq \sup_n \|f_n\|_{L^p} \|g - g_j\|_{L^q} + \left| \int f_n g_j - \int f g_j \right| + \|f\|_{L^p} \|g - g_j\|_{L^q}. \end{aligned}$$

Thus, taking the limit $n \rightarrow \infty$ on both sides, and using the fact that for each j , the second term vanishes under the limit by hypothesis, one has

$$\lim_{n \rightarrow \infty} \left| \int f_n g - \int f g \right| \leq \sup_n \|f_n\|_{L^p} \|g - g_j\|_{L^q} + \|f\|_{L^p} \|g - g_j\|_{L^q}.$$

Taking the limit $j \rightarrow \infty$ shows that $\int f_n g \rightarrow \int f g$ so that f_n converges to f weakly.

- (c) Suppose $1 < p < \infty$, and assume $\sup_n \|f_n\|_{L^p} < \infty$. First, since X was assumed to be separable, then by Exercise 10(b), one has that $L^q(X, \mu)$ is separable. So let $\{g_j\}_{j=1}^\infty \subset L^q(X, \mu)$ be a countable dense collection. We consider the sequence $\{\int f_n g_1 \, d\mu\}_{n=1}^\infty \subset \mathbb{R}$. First, it follows that this sequence is uniformly bounded: for every n ,

$$\left| \int f_n g_1 \, d\mu \right| \leq \int |f_n| |g_1| \, d\mu \leq \|f_n\|_{L^p} \|g_1\|_{L^q} \leq \sup_n \|f_n\|_{L^p} \|g_1\|_{L^q} < \infty.$$

Thus there exists a subsequence $\{f_{n_k}^{(1)}\}_{k=1}^\infty$ so that $\{\int f_{n_k}^{(1)} g_1 \, d\mu\}_{k=1}^\infty$ converges. Now consider the sequence $\{\int f_{n_k}^{(1)} g_2 \, d\mu\}_{k=1}^\infty$. As before, this sequence is uniformly bounded so that there exists a further subsequence $\{f_{n_k}^{(2)}\}_{k=1}^\infty$ such that $\{\int f_{n_k}^{(2)} g_2 \, d\mu\}_{k=1}^\infty$ converges. By construction, $\{\int f_{n_k}^{(2)} g_1 \, d\mu\}_{k=1}^\infty$ also converges. In this manner, for each j , we can obtain a subsequence $\{f_{n_k}^{(j)}\}_{k=1}^\infty$ so that $\{\int f_{n_k}^{(j)} g_\ell \, d\mu\}_{k=1}^\infty$ converges for all $\ell \leq j$. Now construct the sequence $\{f_{n_k}\}_{k=1}^\infty = \{f_{n_k}^{(j)}\}_{j=1}^\infty$. By construction, $\{\int f_{n_k} g_j \, d\mu\}_{k=1}^\infty$ converges for all j . Now we claim that $\{\int f_{n_k} g \, d\mu\}_{k=1}^\infty$ converges for all $g \in L^q$. Let $\varepsilon > 0$ be fixed, $M := \sup_n \|f_n\|_{L^p} < \infty$, j large enough so that $\|g - g_j\|_{L^q} < \varepsilon/(3M)$, and N large enough so that for all $k, m \geq N$, $|\int f_{n_k} g_j - \int f_{n_m} g_j| < \varepsilon/3$. Then for all $k, m \geq N$, one has

$$\begin{aligned} \left| \int f_{n_k} g - \int f_{n_m} g \right| &\leq 2 \sup_n \|f_n\|_{L^p} \cdot \|g - g_j\|_{L^q} + \left| \int f_{n_k} g_j - \int f_{n_m} g_j \right| \\ &< \frac{2\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Thus taking $\varepsilon \rightarrow 0$ concludes the proof of the above claim. Now define the linear functional $T : L^q \rightarrow \mathbb{R}$ as follows: for any $g \in L^q$, let

$$T(g) := \lim_{n_j \rightarrow \infty} \int f_{n_j} g.$$

It is straightforward to see that this functional is bounded. Thus since $1 < p, q < \infty$ and p, q are conjugates, by the Riesz Representation Theorem, there exists a unique $f \in L^p$ so that

$$\int f g = T(g) = \lim_{n_j \rightarrow \infty} \int f_{n_j} g \quad (g \in L^q(X, \mu)).$$

Thus the proof concludes.

Problem 2.2. The following are simple generalizations of the Hausdorff-Young inequalities.

- (a) Suppose $\{\varphi_n\}$ is an orthonormal sequence on $L^2(X, \mu)$. Assume also that $|\varphi_n(x)| \leq M$ for all n . If $a_n = \int f \bar{\varphi}_n d\mu$, then $\|a_n\|_{L^q} \leq M^{(2/p)-1} \|f\|_{L^p(X)}$, $1 \leq p \leq 2$, $1/p + 1/q = 1$.
- (b) Suppose $f \in L^p$ on the torus $\mathbb{T}^d$, and $a_n = \int_{\mathbb{T}^d} f(x) e^{-2\pi i n \cdot x} dx$, $n \in \mathbb{Z}^d$. Then $\|\{a_n\}\|_{L^q(\mathbb{Z}^d)} \leq \|f\|_{L^p(\mathbb{T}^d)}$, where $1/q \leq 1 - 1/p$.

- (a) We initially define the operator T on simple functions f by $Tf = \{a_n\}$, where

$$a_n := \int f \bar{\varphi}_n d\mu.$$

Then clearly, for $(p_0, q_0) = (1, \infty)$, one has that since for each n ,

$$|a_n| \leq \int |f| |\bar{\varphi}_n| d\mu \leq \int M |f| = M \|f\|_{L^1(X)},$$

$\|a_n\|_{L^\infty} = \sup |a_n| \leq M \|f\|_{L^1(X)}$. Thus by Proposition 5.4, we conclude that there exists a unique extension T to all of $L^1(X)$ satisfying the same bound. Likewise, for $(p_1, q_1) = (2, 2)$ and any simple function f , Bessel's inequality implies that

$$\|a_n\|_{L^2} = \left(\sum_{n=1}^{\infty} |a_n|^2 \right)^{1/2} \leq \|f\|_{L^2(X)} = M^{\frac{2}{2}-1} \|f\|_{L^2(X)}.$$

Thus by Proposition 5.4, there exists a unique extension T to all of $L^2(X)$ satisfying the same bound. Then, the Riesz-Thorin Interpolation Theorem tells us that $\|a_n\|_{L^q} \leq M \|f\|_{L^p(X)}$ where

$$\frac{1}{p} = \frac{1-t}{p_0} + \frac{t}{p_1} = 1 - \frac{1}{2}t, \quad \frac{1}{q} = \frac{1-t}{q_0} + \frac{t}{q_1} = \frac{1}{2}t,$$

where $t \in [0, 1]$ (thus corresponding to $1 \leq p \leq 2$, $p^{-1} + q^{-1} = 1$), and where

$$M \leq M^{1-t} = M^{2(1-(1/2)t)-1} = M^{\frac{2}{p}-1}.$$

- (b) As before, initially define the operator T on simple functions f on $\mathbb{T}^d$ by $Tf = \{a_n\}$, where for each $n \in \mathbb{Z}^d$,

$$a_n := \int_{\mathbb{T}^d} f(x) e^{-2\pi i n \cdot x} dx.$$

Clearly for $(p_0, q_0) = (1, \infty)$, one has

$$|a_n| \leq \int_{\mathbb{T}^d} |f(x)| = \|f\|_{L^1(\mathbb{T}^d)},$$

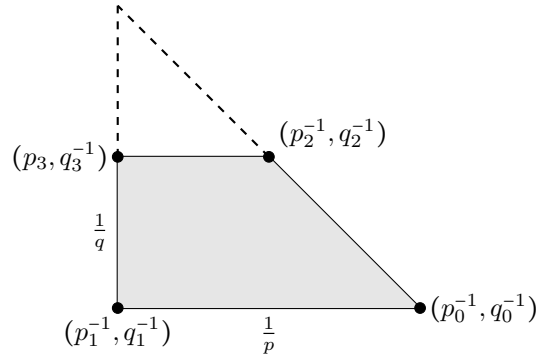
so that

$$\|\{a_n\}\|_{L^\infty(\mathbb{Z}^d)} = \sup_n |a_n| \leq \|f\|_{L^1(\mathbb{T}^d)}.$$

Thus Proposition 5.4 gives a unique extension of T to all $L^1(\mathbb{T}^d)$ that satisfies the same bound. Now consider $(p_1, q_1) = (\infty, \infty)$. Then for any $n \in \mathbb{Z}^d$, one has that

$$|a_n| \leq \int_{\mathbb{T}^d} |f(x)| \leq \|f\|_{L^\infty(\mathbb{T}^d)},$$

where we assume that dx denotes the normalized measure on $\mathbb{T}^d$. Thus taking the supremum over all $n \in \mathbb{Z}^d$, and using Proposition 5.4 concludes the claim. Likewise, one can show this property holds for the pair $(p_2, q_2) = (2, 2)$. Since $\mathbb{T}^d$ has finite measure, it follows that for all $p \geq 2$ (i.e., $1/p \leq 1/2$), $L^p(\mathbb{T}^d) \hookrightarrow L^2(\mathbb{T}^d)$. Thus, using Proposition 5.4, we also get the endpoint $(p_3, q_3) = (p, 2)$ for all $p \geq 2$. The Riesz diagram as of now is given by



Problem 2.3. Check that an inequality of the form $\|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq A \|f\|_{L^p(\mathbb{R}^d)}$ (holding for all simple functions f) is possible if and only if $1/p + 1/q = 1$.
 [Hint: Let $f_r(x) = f(rx)$, $r > 0$. Then $\hat{f}_r(\xi) = \hat{f}(\xi/r)r^{-d}$.]

Let f be an arbitrary fixed simple function. As suggested by the hint, let $r > 0$, and define $f_r(x) = f(rx)$ so that $\hat{f}_r(\xi) = \hat{f}(\xi/r)r^{-d}$. We note that f_r is also a simple function. We now compute,

$$\begin{aligned} \|\hat{f}_r\|_{L^q(\mathbb{R}^d)} &= \left(\int_{\mathbb{R}^d} |\hat{f}(\xi/r)|^q r^{-dq} d\xi \right)^{1/q} \\ &= r^{-d} \left(\int_{\mathbb{R}^d} |\hat{f}(\xi/r)|^q d\xi \right)^{1/q} \\ &= r^{-d} \left(\int_{\mathbb{R}^d} |\hat{f}(\eta)|^q r^d d\eta \right)^{1/q} \\ &= r^{d(-1+1/q)} \|\hat{f}\|_{L^q(\mathbb{R}^d)}. \end{aligned}$$

Likewise,

$$\begin{aligned} \|f_r\|_{L^p(\mathbb{R}^d)} &= \left(\int_{\mathbb{R}^d} |f(rx)|^p dx \right)^{1/p} \\ &= \left(\int_{\mathbb{R}^d} |f(y)| r^{-d} dy \right)^{1/p} = r^{-d/p} \|f\|_{L^p(\mathbb{R}^d)}. \end{aligned}$$

So we have the following: the above estimate holds for all simple functions f if and only if for any $r > 0$,

$$\|\hat{f}_r\|_{L^q(\mathbb{R}^d)} \leq A \|f\|_{L^p(\mathbb{R}^d)} \quad \text{and} \quad \|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq A \|f\|_{L^p(\mathbb{R}^d)}.$$

But our work above shows that the first inequality is equivalent to the estimate

$$r^{-d+d/q} \|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq r^{-d/p} \|f\|_{L^p(\mathbb{R}^d)} \iff \|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq r^{d(1-1/q-1/p)} A \|f\|_{L^p(\mathbb{R}^d)}.$$

Thus since $r > 0$ is arbitrary, the above estimate is consistent with the two inequality we have above if and only if $1 - 1/q - 1/p = 0$ iff $1/q + 1/p = 1$.

Problem 2.10. Suppose $f \in L^p(\mathbb{R})$. Verify that:

- (a) $\|f * \mathcal{P}_y\|_{L^p(\mathbb{R})} \leq \|f\|_{L^p}$, $1 \leq p \leq \infty$.
- (b) $f * \mathcal{P}_y \rightarrow f$, as $y \rightarrow 0$ in the L^p norm when $1 \leq p < \infty$.

(a) Let $\mathcal{P}_y(x)$ denote the **Poisson kernel**,

$$\mathcal{P}_y(x) := \frac{1}{\pi} \frac{y}{x^2 + y^2}.$$

Let $p_0 = 1$, and define the operator T on $L^{p_0}(\mathbb{R})$, initially just on the set of all simple functions, by $Tf := (f * \mathcal{P}_y)(x)$. First, if $y = 0$, then $\mathcal{P}_y(x) \equiv 0$ so that $Tf \equiv 0$, and the result holds trivially. So assume $y \neq 0$ (wlog, assume $y > 0$). Then we compute,

$$\begin{aligned} \|\mathcal{P}_y\|_{L^1(\mathbb{R})} &= \int_{\mathbb{R}} \frac{1}{\pi} \frac{|y|}{x^2 + y^2} dx \\ &= \frac{1}{\pi} \frac{|y|}{y} \arctan\left(\frac{x}{y}\right) \Big|_{-\infty}^{\infty} = \frac{1}{\pi} \frac{|y|}{y} \cdot \left[\frac{\pi}{2} + \frac{\pi}{2}\right] = \frac{|y|}{y} = 1. \end{aligned}$$

Thus for $p_0 = 1$, we have

$$\begin{aligned} \|Tf\|_{L^1(\mathbb{R})} &= \int_{\mathbb{R}} \left| \frac{1}{\pi} \int_{\mathbb{R}} \frac{f(x-t)y}{y^2 + t^2} dt \right| dx \\ &\leq \int_{\mathbb{R}} \int_{\mathbb{R}} |f(x-t)| \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dx dt \\ &= \int_{\mathbb{R}} \|f(\cdot - t)\|_{L^1(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dt \\ &= \int_{\mathbb{R}} \|f\|_{L^1(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dt = \|f\|_{L^1(\mathbb{R})} \|\mathcal{P}_y\|_{L^1(\mathbb{R})} = \|f\|_{L^1(\mathbb{R})}, \end{aligned}$$

where the second line follows from Minkowski's inequality, and the fourth line follows from translation invariance of the L^p norms. Thus by Proposition 5.4, T extends to a unique operator on all of $L^1(\mathbb{R})$ satisfying the same estimate. Now define the operator T on $L^{p_1}(\mathbb{R})$, where $p_1 = \infty$, initially on just the set of all simple functions by $Tf := (f * \mathcal{P}_y)(x)$. As before, the proof is trivial if $y = 0$. So assume $y \neq 0$. Then for each x ,

$$\begin{aligned} |Tf(x)| &= \left| \int_{\mathbb{R}} \frac{1}{\pi} \frac{f(x-t)y}{y^2 + t^2} dt \right| \\ &\leq \int_{\mathbb{R}} |f(x-t)| \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dt \\ &\leq \int_{\mathbb{R}} \|f(x-\cdot)\|_{L^\infty(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dt \\ &= \int_{\mathbb{R}} \|f\|_{L^\infty(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dt = \|f\|_{L^\infty(\mathbb{R})} \cdot \|\mathcal{P}_y\|_{L^1(\mathbb{R})} = \|f\|_{L^\infty(\mathbb{R})}. \end{aligned}$$

Using Proposition 5.4, the result extends to all of $L^\infty(\mathbb{R})$. On the Riesz diagram, we have proven the estimate for the endpoints $(0,0)$ and $(1,1)$. Thus by the Riesz-Thorin interpolation theorem, the theorem follows for all points on the diagonal line connecting these endpoints (i.e., for all $1 \leq p \leq \infty$).

(b) Now let $1 \leq p < \infty$, and let $\delta > 0$ be fixed. By continuity of translations in $L^p(\mathbb{R})$, there exists $\varepsilon > 0$

such that $|t| \leq \delta \implies \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} < \varepsilon/2$. Thus we compute,

$$\begin{aligned} \|f * \mathcal{P}_y - f\|_{L^p} &= \left(\int_{\mathbb{R}} \left| \int_{\mathbb{R}} \frac{1}{\pi} \frac{f(x-t)y}{t^2 + y^2} dt - f(x) \right|^p dx \right)^{1/p} \\ &= \left(\int_{\mathbb{R}} \left| \int_{\mathbb{R}} [f(x-t) - f(x)] \cdot \frac{1}{\pi} \frac{y}{t^2 + y^2} dt \right|^p dx \right)^{1/p} \\ &\leq \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |f(x-t) - f(x)| \cdot \left| \frac{1}{\pi} \frac{y}{t^2 + y^2} \right|^p dx \right)^{1/p} dt \\ &= \int_{\mathbb{R}} \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2 + y^2} \right| dt \\ &= \int_{|t| \leq \delta} \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2 + y^2} \right| dt + \int_{|t| > \delta} \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2 + y^2} \right| dt \\ &\leq \varepsilon/2 + 2 \|f\|_{L^p(\mathbb{R})} \cdot \int_{|t| > \delta} |\mathcal{P}_y(t)| dt, \end{aligned}$$

where the second line follows from $\|\mathcal{P}_y\|_{L^1(\mathbb{R})} = 1$ for any $y \in \mathbb{R}$, and the third line follows from Minkowski's inequality. Since $\mathcal{P}_y(t) \rightarrow 0$ for all $t \neq 0$ in the limit $y \rightarrow 0^+$, it follows that the second term tends to 0 in the limit $y \rightarrow 0^+$. I.e., there exists some y_0 so that for all $y < y_0$, the second term is bounded above by $\varepsilon/(2\|f\|_{L^p(\mathbb{R})})$. Therefore, we conclude that $\|f * \mathcal{P}_{y_0} - f\|_{L^p(\mathbb{R})} \leq \varepsilon$ so that $f * \mathcal{P}_y \rightarrow f$ in the L^p norm.

Problem 2.16. Prove that the space $\mathbf{H}_r^1$ is complete as follows. Let $\{f_n\}$ be a Cauchy sequence in $\mathbf{H}_r^1$. Then since $\{f_n\}$ is also Cauchy in L^1 , there is an L^1 function f so that $f = \lim_{n \rightarrow \infty} f_n$ in the L^1 -norm. Now for an appropriate subsequence $\{n_k\}$, write $f = f_{n_1} + \sum_{k=1}^{\infty} (f_{n_{k+1}} - f_{n_k})$.

Let $\mathbf{H}_r^1$ denote the Hardy space, and let $\{f_n\} \subset \mathbf{H}_r^1$ be a Cauchy sequence. First, we begin by showing that each $f \in L^1$. Let $\sum_j \lambda_j \mathbf{a}_j$ be an arbitrary decomposition of f , where each $\mathbf{a}_j$ is an atom, and $\sum |\lambda_j| < \infty$. Thus since the sum converges under the L^1 norm, Fubini/Tonelli implies that

$$\int_{\mathbb{R}} |f| dx = \int_{\mathbb{R}} \left| \sum_1^{\infty} \lambda_k \mathbf{a}_k \right| \leq \int_{\mathbb{R}} \sum_1^{\infty} |\lambda_k| |\mathbf{a}_k| = \sum_1^{\infty} \int_{\mathbb{R}} |\lambda_k| |\mathbf{a}_k| = \sum_1^{\infty} |\lambda_k| < \infty.$$

Thus since $\sum \lambda_j \mathbf{a}_j$ was an arbitrary decomposition of f into a sum of atoms, it follows that $f \in L^1$. In particular, taking the infimum over all such decompositions also yields the estimate $\|f\|_{L^1} \leq \|f\|_{\mathbf{H}_r^1}$. Thus if $\{f_n\}$ is Cauchy in $\mathbf{H}_r^1$, they are also Cauchy in L^1 . Since the latter is a Banach space, there exists an L^1 function f such that $f = \lim_{n \rightarrow \infty} f_n$ in the L^1 -norm. But this implies there exists a subsequence $\{f_{n_k}\}$ that converges to f pointwise a.e. Fix $\varepsilon > 0$. Since $\{f_n\}$ is Cauchy in $\mathbf{H}_r^1$, for every $k \geq 1$, there exists a positive integer N_k such that for all $m, n > N_k$,

$$\|f_m - f_n\|_{\mathbf{H}_r^1} < \frac{\varepsilon}{2^k}.$$

Thus from the subsequence $\{f_{n_k}\}_{k=1}^{\infty}$, we construct another subsequence as follows: we choose $n_1 > N_1$. Next, we choose $n_2 > \max\{n_1, N_2\}$. Repeating inductively, we obtain a subsequence $\{f_{n_k}\}$ such that

$$\sum_{k=1}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|_{\mathbf{H}_r^1} \leq \sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon < \infty.$$

Now we recall that

$$f = \lim_{j \rightarrow \infty} f_{n_j} = f_{n_1} + \lim_{j \rightarrow \infty} \sum_{k=1}^j (f_{n_{k+1}} - f_{n_k}) = f_{n_1} + \sum_{k=1}^{\infty} (f_{n_{k+1}} - f_{n_k}).$$

Thus

$$\|f\|_{\mathbf{H}_r^1} \leq \|f_{n_1}\|_{\mathbf{H}_r^1} + \sum_{k=1}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|_{\mathbf{H}_r^1} < \infty,$$

which establishes that $f \in \mathbf{H}_r^1$. All that remains to be shown is that $f_n \rightarrow f$ under the $\mathbf{H}_r^1$ norm. We compute,

$$\|f - f_n\|_{\mathbf{H}_r^1} \leq \|f - f_{n_k}\|_{\mathbf{H}_r^1} + \|f_{n_k} - f_n\|_{\mathbf{H}_r^1}.$$

Since the sequence $\{f_n\}$ is Cauchy in $\mathbf{H}_r^1$, the second term vanishes in the limit $n, k \rightarrow \infty$. On the other hand, since

$$\|f - f_{n_k}\|_{\mathbf{H}_r^1} \leq \sum_{j=k}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|_{\mathbf{H}_r^1}.$$

By our construction of the subsequence $\{f_{n_k}\}$, we can control the sum by ε . Thus we conclude that $f_n \rightarrow f$ under the $\mathbf{H}_r^1$ norm, which means that $\mathbf{H}_r^1$ is a Banach space.

Problem 2.20. When $f \in L^1(\mathbb{R}^d)$ we know that its Fourier transform is bounded and $\hat{f}(\xi)$ tends to 0 as $|\xi| \rightarrow \infty$ (the Riemann-Lebesgue lemma), but no better assertion about the “smallness” of $\hat{f}$ can be made. (For the analogous result for Fourier series, see Chapter 3 in Book III.) Show, however, that for $f \in \mathbf{H}_r^1$ we have

$$\int_{\mathbb{R}^d} |\hat{f}(\xi)| \frac{d\xi}{|\xi|^d} \leq A \|f\|_{\mathbf{H}_r^1}.$$

[Hint: Verify this for atoms.]

First, assume that for every atom $\mathbf{a}$, the above estimate holds. Then for any decomposition, $f = \sum_{k=1}^{\infty} \lambda_k \mathbf{a}_k$ into atoms,

$$\begin{aligned} \int_{\mathbb{R}^d} |\hat{f}(\xi)| \frac{d\xi}{|\xi|^d} &= \int_{\mathbb{R}^d} \left| \sum_1^{\infty} \lambda_k \hat{\mathbf{a}}_k \right| \frac{d\xi}{|\xi|^d} \\ &\leq \sum_1^{\infty} \int_{\mathbb{R}^d} |\lambda_k| |\hat{\mathbf{a}}_k| \frac{d\xi}{|\xi|^d} \\ &\leq A \sum_1^{\infty} |\lambda_k|. \end{aligned}$$

Thus taking the infimum over all such decompositions of f , we obtain,

$$\int_{\mathbb{R}^d} |\hat{f}(\xi)| \frac{d\xi}{|\xi|^d} \lesssim \|f\|_{\mathbf{H}_r^1}.$$

So now, we will show that the result holds true for every atom. Let $\mathbf{a}$ be an atom on $\mathbb{R}^d$; without loss of generality, we can assume that $\mathbf{a}$ is supported on the unit ball $B_1(0) \subset \mathbb{R}^d$. Assume $\mathbf{a}$ is not identically zero. Then one has that

$$\hat{\mathbf{a}}(\xi) := \int_{\mathbb{R}^d} \mathbf{a}(x) e^{-2\pi i x \cdot \xi} dx = \int_{|x| \leq 1} \mathbf{a}(x) e^{-2\pi i x \cdot \xi} dx = \int_{|x| \leq 1} \mathbf{a}(x) (e^{-2\pi i x \cdot \xi} - 1) dx,$$

where the final equality follows from the fact that $\int_{\mathbb{R}^d} \mathbf{a}(x) dx = 0$. Thus,

$$\begin{aligned} \int_{\mathbb{R}^d} |\hat{\mathbf{a}}(\xi)| \frac{d\xi}{|\xi|^d} &= \int_{|\xi| \leq 1} \left| \int_{|x| \leq 1} \mathbf{a}(x) \frac{e^{-2\pi i x \cdot \xi} - 1}{|\xi|^d} dx \right| d\xi + \int_{|\xi| > 1} |\hat{\mathbf{a}}(\xi)| \frac{d\xi}{|\xi|^d} \\ &\leq \int_{|x| \leq 1} \int_{|\xi| \leq 1} |\mathbf{a}(x)| \cdot \left| \frac{e^{-2\pi i x \cdot \xi} - 1}{|\xi|^d} \right| d\xi dx + \int_{|\xi| > 1} |\hat{\mathbf{a}}(\xi)| \frac{d\xi}{|\xi|^d} \\ &\leq 2\pi \|\mathbf{a}\|_{L^1(\mathbb{R}^d)} + \left(\int_{|\xi| > 1} |\hat{\mathbf{a}}(\xi)|^2 d\xi \right)^{1/2} \left(\int_{|\xi| > 1} \frac{1}{|\xi|^{2d}} d\xi \right)^{1/2} \\ &= 2\pi \|\mathbf{a}\|_{L^1(\mathbb{R}^d)} + \left(\int_{|x| \leq 1} |\mathbf{a}(x)|^2 dx \right)^{1/2} \\ &\leq 2\pi \|\mathbf{a}\|_{L^1(\mathbb{R}^d)} + \left(\frac{1}{m(B_1(0))} \right)^{1/2} \leq 2\pi + \frac{1}{m(B_1(0))^{1/2}} \equiv A < \infty. \end{aligned}$$

2.2 Evans

Problem 2.18. (Stokes' rule) Assume u solves the initial-value problem

$$\begin{cases} u_{tt} - \Delta u = 0 & \text{in } \mathbb{R}^n \times (0, \infty) \\ u = 0, u_t = h, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

Show that $v := u_t$ solves

$$\begin{cases} v_{tt} - \Delta v = 0 & \text{in } \mathbb{R}^n \times (0, \infty) \\ v = h, v_t = 0, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

This is *Stokes' rule*.

Assume u solves the given IVP, and let $v := u_t$. One immediately notes that on $\mathbb{R}^n \times \{t = 0\}$, since $u_t = h$, $v = h$. Second, $v_t = u_{tt}$. On $\mathbb{R}^n \times \{t = 0\}$, $u \equiv 0$ so that $\Delta u = 0$. But since u satisfies the wave equation $u_{tt} = \Delta u \equiv 0$. Thus $v_t \equiv 0$ on $\mathbb{R}^n \times \{t = 0\}$. Next we note that ∂_t and ∂_{x_j} commute for any $1 \leq j \leq n$. Thus since u satisfies $u_{tt} - \Delta u = 0$, differentiating both sides by t , one obtains

$$u_{ttt} - \Delta u_t = 0 \iff v_{tt} - \Delta v = 0$$

in $\mathbb{R}^n \times (0, \infty)$.

Problem 2.19.

(a) Show the general solution of the PDE $u_{xy} = 0$ is

$$u(x, y) = F(x) + G(y)$$

for arbitrary functions F, G .

(b) Using the change of variables $\xi = x + t$, $\eta = x - t$, show $u_{tt} - u_{xx} = 0$ if and only if $u_{\xi\eta} = 0$.

(c) Use (a) and (b) to rederive d'Alembert's formula.

(d) Under what conditions on the initial data g, h is the solution u a right-moving wave? A left-moving wave?

(a) Consider the PDE $u_{xy} = 0$. Let $v(x, y) = u_x$. Then v satisfies the differential equation $v_y = 0$. Thus since v is independent of y , one has $u_x(x, y) = v(x, y) = f(x)$ for some arbitrary function f . But then integrating both sides with respect to x , and supposing that $F(x)$ is the antiderivative of $f(x)$,

$$u(x, y) = F(x) + G(y),$$

where G is some arbitrary function of y .

(b) Define $\xi = x + t$ and $\eta = x - t$ so that $x = (\xi + \eta)/2$ and $t = (\xi - \eta)/2$. Then,

$$\begin{aligned} u_\xi &= u_x \frac{\partial x}{\partial \xi} + u_t \frac{\partial t}{\partial \xi} \\ &= \frac{1}{2} u_x + \frac{1}{2} u_t. \\ u_{\xi\eta} &= \frac{1}{2} \left[u_{xx} \frac{\partial x}{\partial \eta} + u_{xt} \frac{\partial t}{\partial \eta} \right] + \frac{1}{2} \left[u_{tx} \frac{\partial x}{\partial \eta} + u_{tt} \frac{\partial t}{\partial \eta} \right] \\ &= \frac{1}{2} [u_{xx} - u_{xt}] + \frac{1}{2} [u_{tx} - u_{tt}] = \frac{1}{2} [u_{tt} - u_{xx}], \end{aligned}$$

where we used the fact that $u_{xt} = u_{tx}$. Thus, the proof concludes.

(c) Consider the initial value problem,

$$\begin{cases} u_{tt} - u_{xx} = 0 & (\text{in } \mathbb{R} \times (0, \infty)) \\ u = g, u_t = h & (\text{on } \mathbb{R} \times \{t = 0\}). \end{cases}$$

Define the change of variables $\xi = x + t$ and $\eta = x - t$ as above. By (b), $u_{tt} - u_{xx} = 0$ if and only if $u_{\xi\eta} = 0$. By (a), it follows that u must be of the form,

$$u(\xi, \eta) = F(\xi) + G(\eta).$$

Thus, $u(x, t) = F(x + t) + G(x - t)$. Now we must match the initial conditions. At $t = 0$,

$$\begin{aligned} u(x, 0) &= F(x) + G(x) = g(x). \\ u_t(x, 0) &= F'(x) - G'(x) = h(x). \end{aligned}$$

Differentiating the first equation, one first obtains the system

$$\begin{cases} F'(x) + G'(x) = g'(x) \\ F'(x) - G'(x) = h(x). \end{cases}$$

Thus, we get

$$F'(x) = \frac{1}{2}(g'(x) + h(x)) \implies F(x + t) = \frac{1}{2}[g(x + t) - g(0)] + \frac{1}{2} \int_0^{x+t} h(y) dy + F(0).$$

Likewise, we get

$$G(x - t) = \frac{1}{2}[g(x - t) - g(0)] - \int_0^{x-t} h(y) dy + G(0).$$

Adding the two equations, and using the fact that $F(0) + G(0) = g(0)$, we conclude that

$$u(x, t) = \frac{1}{2}[g(x - t) + g(x + t)] + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy,$$

which is precisely d'Alembert's formula.

- (d) To get a left-moving wave, we want the functions of $(x - t)$ to cancel out. Suppose $h(x) = g'(x)$. Then the fundamental theorem of calculus implies that

$$u(x, t) = \frac{1}{2}[g(x - t) + g(x + t)] + \frac{1}{2} \int_{x-t}^{x+t} g'(y) dy = \frac{1}{2}[g(x - t) + g(x + t)] + \frac{1}{2}[g(x + t) - g(x - t)] = g(x + t).$$

Likewise, to get a right-moving wave, we want the functions of $(x + t)$ to cancel out, so we can take $h(x) = -g'(x)$.

Problem 2.21. (Equipartition of Energy). Let $u \in C^2(\mathbb{R} \times [0, \infty))$ solve the initial-value problem for the wave equation in one-dimension:

$$\begin{cases} u_{tt} - u_{xx} = 0 & (\text{in } \mathbb{R} \times (0, \infty)) \\ u = g, u_t = h & (\text{on } \mathbb{R} \times \{t = 0\}). \end{cases}$$

Suppose g, h have compact support. The *kinetic energy* is $k(t) := \frac{1}{2} \int_{-\infty}^{\infty} u_t^2(x, t) dx$ and the *potential energy* is $p(t) := \frac{1}{2} \int_{-\infty}^{\infty} u_x^2(x, t) dx$. Prove

- (i) $k(t) + p(t)$ is constant in t ,
- (ii) $k(t) = p(t)$ for all large enough times t .

- (i) Let us define the kinetic and potential energy, $k(t)$ and $p(t)$ as given above. We wish to show that the total energy

$$e(t) = k(t) + p(t) = \frac{1}{2} \int_{-\infty}^{\infty} u_t^2 + u_x^2 dx$$

is conserved, where u is given by d'Alembert's formula,

$$u(x, t) = \frac{1}{2}[g(x + t) + g(x - t)] + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy.$$

so that

$$\begin{aligned} u_t(x, t) &= \frac{1}{2}[g'(x+t) - g'(x-t) + h(x+t) + h(x-t)], \\ u_x(x, t) &= \frac{1}{2}[g'(x+t) + g'(x-t) + h(x+t) - h(x-t)]. \end{aligned}$$

Let $T > 0$, and let $K \subset \mathbb{R}$ be a compact set large enough so that $\text{supp } g, \text{supp } h$ are both contained in K . Then it follows that u_t^2 and u_x^2 are both supported in $K_T := K + [-T, T]$ for all $0 \leq t \leq T$. Since u_t^2 and u_x^2 are continuous on $\mathbb{R}$ (and thus on K_T) for all $0 \leq t \leq T$, by the extreme value theorem, there exist finite constants M and N so that for all $x \in K$ and $0 \leq t \leq T$, $|u_t|^2 \leq M\chi_{K_T}$ and $|u_x|^2 \leq N\chi_{K_T}$, where both dominating functions are integrable since K_T is compact. Thus using the Dominated Convergence Theorem, taking the derivative of $e(t)$ gives us,

$$\begin{aligned} \dot{e}(t) &= \frac{1}{2} \frac{\partial}{\partial t} \int_{-\infty}^{\infty} u_t^2 + u_x^2 dx = \frac{1}{2} \frac{\partial}{\partial t} \int_{K_T} u_t^2 + u_x^2 dx \\ &= \frac{1}{2} \int_{K_T} \partial_t(u_t^2 + u_x^2) dx \\ &= \int_{K_T} u_t u_{tt} + u_x \cdot \partial_x u_t dx \\ &= \int_{K_T} u_t(u_{tt} - u_{xx}) dx \quad (\text{by integration by parts}) \\ &= 0, \end{aligned}$$

where there is no boundary term since u_x, u_t are supported inside K_T . So we have shown that $\dot{e}(t) = 0$ for all $0 \leq t \leq T$. Since $T > 0$ was chosen arbitrarily, it follows that $\dot{e}(t) \equiv 0$. Therefore, the total energy is constant in t .

(ii) What we wish is that in the limit $t \rightarrow \infty$,

$$\left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| \rightarrow 0.$$

First, we compute

$$u_t^2 - u_x^2 = h(x+t)h(x-t) - h(x+t)g'(x-t) + h(x-t)g'(x+t) - g'(x-t)g'(x+t).$$

Then

$$\begin{aligned} \left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| &\leq \int_{-\infty}^{\infty} |h(y)h(y-2t)| dy + \int_{-\infty}^{\infty} |h(y)g'(y-2t)| dy \\ &\quad + \int_{-\infty}^{\infty} |h(y-2t)g'(y)| dy + \int_{-\infty}^{\infty} |g'(y-2t)g'(y)| dy. \end{aligned}$$

Let $K \subset \mathbb{R}$ be a compact set large enough that contains the support of both g and h . Then we may write,

$$\begin{aligned} \left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| &\leq \int_K |h(y)h(y-2t)| dy + \int_K |h(y)g'(y-2t)| dy \\ &\quad + \int_K |h(y-2t)g'(y)| dy + \int_K |g'(y-2t)g'(y)| dy. \end{aligned}$$

We will only show that the first term on the right vanishes since the work for the remaining terms is identical. Since $h(u) \rightarrow 0$ as $|u| \rightarrow \infty$ (by compact support of h), we know that pointwise $|h(y)h(y-2t)| \rightarrow 0$ as $t \rightarrow \infty$. Next since h is continuous, there exists some finite $M > 0$ so that $|h(y)|, |h(y-2t)| \leq M$ for all $y \in K$. Thus, $|h(y)h(y-2t)| \leq M^2\chi_K \in L^1(\mathbb{R})$ since K is compact. Hence, by the Dominated Convergence Theorem, one has that the first term vanishes as $t \rightarrow \infty$. As we stated before, the remaining three terms also vanish as $t \rightarrow \infty$. Thus, we conclude that as $t \rightarrow \infty$,

$$\left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| \rightarrow 0.$$

2.3 Luk Notes

Problem N.1. (Luk, pg. 3) The Einstein vacuum equations describe the propagation of gravitational waves in the absence of matter and take the form $\text{Ric}(g) = 0$, where the Lorentzian metric g is the unknown. Show that in a coordinate system, these equations take the form

$$\text{Ric}(g)_{\mu\nu} = -\frac{1}{2}g^{\alpha\beta}\partial_{\alpha\beta}^2 g_{\mu\nu} - \frac{1}{2}g^{\alpha\beta}\partial_{\mu\nu}^2 g_{\alpha\beta} + \frac{1}{2}g^{\alpha\beta}\partial_{\alpha\nu}^2 g_{\beta\mu} + \frac{1}{2}g^{\alpha\beta}\partial_{\beta\mu}^2 g_{\alpha\nu} + F_{\mu\nu}(g, \partial g),$$

where $F_{\mu\nu}(g, \partial g)$ is a function of g and its derivatives.

Let (M, g) be a Lorentzian manifold, and let $\text{Ric}(g)$ denote the Ricci curvature tensor on M . Let $(x^j)_{j=1}^n$ denote a local coordinate system, and let $(\partial_j)_{j=1}^n$ be the corresponding vector fields. Let R denote the Riemann curvature tensor. Then if R is given in coordinates by $R = R_{ijk}{}^\ell dx^i \otimes dx^j \otimes dx^k \otimes \partial_\ell$, then

$$R_{ijk}{}^\ell = \partial_i \Gamma_{jk}^\ell - \partial_j \Gamma_{ik}^\ell + \Gamma_{jk}^m \Gamma_{im}^\ell - \Gamma_{ik}^m \Gamma_{jm}^\ell,$$

where the $\Gamma_{**}^\bullet$ are the *Christoffel symbols*. One can also show that in coordinates,

$$\Gamma_{jk}^i = \frac{1}{2}g^{im}(\partial_k g_{mj} + \partial_j g_{mk} - \partial_m g_{jk}).$$

Thus, we then compute,

$$R_{ijk}{}^\ell = \frac{1}{2}g^{\ell m}(\partial_{ki}^2 g_{mj} + \partial_{ji}^2 g_{mk} - \partial_{mi}^2 g_{jk}) + \{\dots\} - \frac{1}{2}g^{\ell m}(\partial_{kj}^2 g_{mi} + \partial_{ij}^2 g_{mk} - \partial_{mj}^2 g_{ik}) - \{\dots\} + \{\dots\}.$$

Then the component functions of the Ricci curvature tensor in coordinates are,

$$\text{Ric}(g)_{ij} = R_{\ell ij}{}^\ell = -\frac{1}{2}g^{\ell m}\partial_{\ell m}^2 g_{ij} - \frac{1}{2}g^{\ell m}\partial_{ij}^2 g_{\ell m} + \frac{1}{2}g^{\ell m}\partial_{im}^2 g_{\ell j} + \frac{1}{2}g^{\ell m}\partial_{\ell j}^2 g_{im} + F_{ij}(g, \partial g).$$

Thus to agree with the format presented in the problem, let $\ell = \alpha$, $m = \beta$, $i = \mu$, and $j = \nu$.

Problem N.2. (Luk, pg. 18) Let ϕ be a solution to the Cauchy problem,

$$\begin{cases} \partial_\alpha(a^{\alpha\beta}\partial_\beta\phi) = F \\ (\phi, \partial_t\phi)|_{\{t=0\}} = (\phi_0, \phi_1). \end{cases}$$

Show that

$$\sup_{t \in [0, T]} \|\phi\|_{L^2(\mathbb{R}^n)}(t) \leq C \left(\|\phi_0\|_{L^2(\mathbb{R}^n)} + \int_0^T \|\partial_t\phi\|_{L^2(\mathbb{R}^n)}(t) dt \right).$$

Let ϕ be a sufficiently regular solution to the aforementioned Cauchy problem. Then the fundamental theorem of calculus tells us that

$$\phi(x, t) = \phi_0(x) + \int_0^t \partial_t\phi(t', x) dt',$$

for any $t \in [0, T]$. Thus, we compute,

$$\begin{aligned} |\phi(x, t)|^2 &\leq |\phi_0|^2 + \left| \int_0^t \partial_t\phi(t', x) dt' \right|^2 \\ \implies \|\phi\|_{L^2(\mathbb{R}^n)}^2 &\leq 2\|\phi_0\|_{L^2(\mathbb{R}^n)}^2 + 2 \int_{\mathbb{R}^n} \left| \int_0^t \partial_t\phi dt' \right|^2 dx \\ \implies \|\phi\|_{L^2(\mathbb{R}^n)} &\leq 2^{1/2} \|\phi_0\|_{L^2(\mathbb{R}^n)} + 2^{1/2} \left(\int_{\mathbb{R}^n} \left| \int_0^t \partial_t\phi dt' \right|^2 dx \right)^{1/2} \\ &\leq 2^{1/2} \|\phi_0\|_{L^2(\mathbb{R}^n)} + 2^{1/2} \int_0^t \|\partial_t\phi\|_{L^2(\mathbb{R}^n)} dt' \\ &\leq 2^{1/2} \left(\|\phi_0\|_{L^2(\mathbb{R}^n)} + \int_0^T \|\partial_t\phi\|_{L^2(\mathbb{R}^n)} dt \right), \end{aligned}$$

where the fourth line follows the Minkowski inequality, and the final line follows from the fact that $\|\partial_t \phi\|_{L^2(\mathbb{R}^n)} \geq 0$.

Problem N.3. Consider the general functional inequality,

$$\|\eta\|_{H^s(\mathbb{R}^n)} \leq C \|\eta\|_{H^{s_1}(\mathbb{R}^n)}^{\theta_1} \|\eta\|_{H^{s_2}(\mathbb{R}^n)}^{\theta_2}$$

for some $C = C(s_1, s_2, s, n) > 0$, where $0 \leq s_1 \leq s \leq s_2$, $\theta_1 + \theta_2 = 1$, and $\theta_1 s_1 + \theta_2 s_2 = s$. Prove this functional inequality by taking the Fourier transform and using Hölder's inequality.

Let $\eta \in H^s(\mathbb{R}^n)$. We recall that

$$\|\eta\|_{H^s(\mathbb{R}^n)}^2 := \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\hat{\eta}|^2 d\xi$$

Let $0 \leq s_1 \leq s \leq s_2$, and let θ_1, θ_2 such that $\theta_1 + \theta_2 = 1$ and $\theta_1 s_1 + \theta_2 s_2 = s$. Then we can write,

$$\begin{aligned} \|\eta\|_{H^s(\mathbb{R}^n)}^2 &= \int_{\mathbb{R}^n} (1 + |\xi|^2)^{\theta_1 s_1} |\hat{\eta}|^{2\theta_1} (1 + |\xi|^2)^{\theta_2 s_2} |\hat{\eta}|^{2\theta_2} d\xi \\ &\leq \left(\int_{\mathbb{R}^n} (1 + |\xi|^2)^{s_1} |\hat{\eta}|^2 d\xi \right)^{\theta_1} \left(\int_{\mathbb{R}^n} (1 + |\xi|^2)^{s_2} |\hat{\eta}|^2 d\xi \right)^{\theta_2} \\ &= \|\eta\|_{H^{s_1}(\mathbb{R}^n)}^{2\theta_1} \|\eta\|_{H^{s_2}(\mathbb{R}^n)}^{2\theta_2}. \end{aligned}$$

Problem E.1.5. Given smooth and compactly supported initial data to the linear wave equation in any dimension, use the Fourier representation formula to show that the $\dot{H}^k$ norm is uniformly bounded in time for any $k \geq 1$.

Let $(\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = (\phi_0, \phi_1)$ be smooth and compactly supported initial data to the linear wave equation in $\mathbb{R}^{1+n}$, and let ϕ be the unique solution to the Cauchy wave problem. In the notes, we explicitly showed that

$$\hat{\phi}(t, \xi) = \hat{\phi}_0 \cos(2\pi t|\xi|) + \frac{\hat{\phi}_1}{2\pi|\xi|} \sin(2\pi t|\xi|).$$

We also recall that the norms for the spaces $\dot{H}^k$ (i.e., the homogeneous Sobolev spaces $\dot{W}^{k,2}$) for $1 \leq k < \infty$ are given by

$$\|u\|_{\dot{H}^k(\mathbb{R}^n)} := \left(\int_{\mathbb{R}^n} |\xi|^{2k} |\hat{u}|^2 d\xi \right)^{1/2}.$$

Thus, we compute,

$$\begin{aligned} \|\phi\|_{\dot{H}^k(\mathbb{R}^n)}^2(t) &= \int_{\mathbb{R}^n} |\xi|^{2k} \left| \hat{\phi}_0 \cos(2\pi t|\xi|) + \frac{\hat{\phi}_1}{2\pi|\xi|} \sin(2\pi t|\xi|) \right|^2 d\xi \\ &\leq 2 \int_{\mathbb{R}^n} |\xi|^{2k} |\hat{\phi}_0|^2 \cos^2(2\pi t|\xi|) d\xi + \frac{1}{2\pi^2} \int_{\mathbb{R}^n} |\xi|^{2k-2} |\hat{\phi}_1|^2 \sin^2(2\pi t|\xi|) d\xi \\ &\leq 2 \|\phi_0\|_{\dot{H}^k(\mathbb{R}^n)}^2 + \frac{1}{2\pi^2} \|\phi_1\|_{\dot{H}^{k-1}(\mathbb{R}^n)}^2, \end{aligned}$$

for all $t \geq 0$. Thus, we conclude that the $\dot{H}^k$ is uniformly bounded in time for any $1 \leq k < \infty$.

Problem E.1.6. (Decay for Klein-Gordon equation) Suppose ϕ satisfies the Klein-Gordon equation; i.e.,

$$\square\phi - \phi = 0$$

on $\mathbb{R} \times \mathbb{R}^n$ with initial data given by $(\phi, \partial_t \phi)|_{t=0} = (\phi_0, \phi_1) \in C_c^\infty \times C_c^\infty$.

- (a) First, show using the Fourier transform that if the solution is sufficiently regular, it is given by the formula

$$\hat{\phi}(t, \xi) = \hat{\phi}_0(\xi) \cos(\sqrt{4\pi^2|\xi|^2 + 1}t) + \frac{\hat{\phi}_1(\xi)}{\sqrt{4\pi^2|\xi|^2 + 1}} \sin(\sqrt{4\pi^2|\xi|^2 + 1}t).$$

- (b) For the $n = 1$ case, prove that $(\sup_x |\phi|)(t) \leq \frac{C}{(1+t)^{\frac{1}{2}}}$, where $C = C(\phi_0, \phi_1) > 0$ is independent of t .

Note that it suffices (why?) to show that

$$\left| \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x \cdot \xi}{t} + \sqrt{4\pi^2\xi^2 + 1}\right)\right) \hat{\phi}_0(\xi); d\xi \right| \leq \frac{C}{t^{1/2}}$$

for t large. Let

$$\varphi := \frac{2\pi x \xi}{t} + \sqrt{4\pi^2|\xi|^2 + 1}.$$

Show that

$$\frac{\partial}{\partial \xi} \varphi = 0 \implies \xi^2 = \frac{x^2}{4\pi^2(t^2 - x^2)}.$$

Show also that

$$\frac{\partial^2}{\partial \xi^2} \varphi \geq \frac{4\pi^2}{(4\pi^2\xi^2 + 1)^{\frac{3}{2}}}.$$

Now split the integral into $\int_{|\xi^2 - \frac{x^2}{4\pi^2(t^2 - x^2)}| \leq \delta} + \int_{|\xi^2 - \frac{x^2}{4\pi^2(t^2 - x^2)}| \geq \delta}$, estimate each part with the above observations and optimized in δ to obtain the desired decay result.

- (c) Finally, can you prove that

$$|\phi| \leq \frac{C}{(1+t)^{\frac{n}{2}}}$$

holds for some $C = C(\phi_0, \phi_1) > 0$ for arbitrary dimensions?

- (a) Consider the Klein-Gordon equation $\square\phi - \phi = 0$ on $\mathbb{R} \times \mathbb{R}^n$, with compactly supported smooth initial data. Assume that a solution exists and that it is sufficiently regular. Then taking the Fourier transform of the equation, one obtains

$$-\partial_t^2 \hat{\phi} - (4\pi^2|\xi|^2 + 1)\hat{\phi} = 0 \implies \partial_t^2 \hat{\phi} = (4\pi^2|\xi|^2 + 1)\hat{\phi}.$$

Thus solving the ODE and using the initial data, we obtain (as desired)

$$\hat{\phi}(t, \xi) = \hat{\phi}_0 \cos(\sqrt{4\pi^2|\xi|^2 + 1}t) + \frac{\hat{\phi}_1}{\sqrt{4\pi^2|\xi|^2 + 1}} \sin(\sqrt{4\pi^2|\xi|^2 + 1}t).$$

- (b) Rewriting the formula for $\hat{\phi}$ (from (a)) in a more convenient way, we obtain

$$\hat{\phi}(t, \xi) = \exp\left(i\sqrt{4\pi^2\xi^2 + 1}t\right) \left(\frac{\hat{\phi}_0}{2} + \frac{\hat{\phi}_1}{2i\sqrt{4\pi^2\xi^2 + 1}}\right) + \exp\left(-i\sqrt{4\pi^2\xi^2 + 1}t\right) \left(\frac{\hat{\phi}_0}{2} - \frac{\hat{\phi}_1}{2i\sqrt{4\pi^2\xi^2 + 1}}\right).$$

Thus taking the inverse Fourier transform,

$$\phi(t, x) = I_1 + I_2 + I_3 + I_4,$$

where

$$\begin{aligned} I_1 &= \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x\xi}{t} + \sqrt{4\pi^2\xi^2 + 1}\right)\right) \frac{\hat{\phi}_0}{2} d\xi & I_2 &= \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x\xi}{t} + \sqrt{4\pi^2\xi^2 + 1}\right)\right) \frac{\hat{\phi}_1}{2i\sqrt{4\pi^2\xi^2 + 1}} d\xi \\ I_3 &= \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x\xi}{t} - \sqrt{4\pi^2\xi^2 + 1}\right)\right) \frac{\hat{\phi}_0}{2} d\xi & I_4 &= \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x\xi}{t} - \sqrt{4\pi^2\xi^2 + 1}\right)\right) \frac{\hat{\phi}_1}{2i\sqrt{4\pi^2\xi^2 + 1}} d\xi. \end{aligned}$$

As suggested in the problem, it suffices to consider just $t \geq 1$. Since ϕ_0, ϕ_1 are compactly supported smooth functions, their Fourier transforms are Schwartz (thus integrable). This means there exist $M_j > 0$, for $j = 1, 2, 3, 4$ such that $|I_j| \leq M_j$. Thus, taking $M > M_1 + \dots + M_4$, we have that for all $x \in \mathbb{R}$, $|\phi| \leq M$. Taking $C > 0$ large enough so that $C \geq M \cdot 2^{1/2}$, we have $|\phi(t, x)| \leq C/(1+t)^{1/2}$ for all $t \geq 0$ and $x \in \mathbb{R}$ so that the result follows. Thus we focus on the case for $t \geq 1$. As in the lecture notes, we also only focus on showing the bound for I_1 . The bound for I_3 will follow nearly identically. As hinted in the problem, define

$$\varphi = \frac{2\pi x\xi}{t} + \sqrt{4\pi^2\xi^2 + 1},$$

and assume that $\partial\varphi/\partial\xi = 0$. I.e.,

$$0 = \frac{\partial\varphi}{\partial\xi} = \frac{2\pi x}{t} + \frac{1}{2}(4\pi^2\xi^2 + 1)^{-1/2} \cdot 8\pi^2\xi \implies \frac{\pi^2 x^2}{t^2} = \frac{4\pi^4\xi^2}{4\pi^2\xi^2 + 1} \implies \xi^2 = \frac{x^2}{4\pi^2(t^2 - x^2)}.$$

Moreover, we compute,

$$\begin{aligned} \partial_\xi^2 \varphi &= \frac{4\pi^2}{\sqrt{4\pi^2\xi^2 + 1}} - \frac{4\pi^2\xi}{(4\pi^2\xi^2 + 1)^{3/2}} \cdot 8\pi^2\xi = \frac{4\pi^2}{\sqrt{4\pi^2\xi^2 + 1}} - \frac{16\pi^4\xi^2}{(4\pi^2\xi^2 + 1)^{3/2}} \\ &= \frac{4\pi^2}{(4\pi^2\xi^2 + 1)^{3/2}}. \end{aligned}$$

Now we consider the integral

$$I_1 = \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x \cdot \xi}{t} + \sqrt{4\pi^2\xi^2 + 1}\right)\right) \hat{\phi}_0(\xi) d\xi = \int_{\mathbb{R}} e^{i\varphi t} \hat{\phi}_0(\xi) d\xi.$$

Taylor expanding φ around ξ_0 (where $\xi_0^2 = x^2/(4\pi(t^2 - x^2))$), we have

$$\varphi = \left(\frac{2\pi x\xi_0}{t} + \sqrt{4\pi^2\xi_0^2 + 1}\right) + \frac{2\pi^2}{(4\pi^2\xi_0^2 + 1)^{3/2}}(\xi - \xi_0)^2 + \dots$$

Therefore,

$$I_1 \approx \hat{\phi}_0(\xi_0) \exp\left(it \cdot \left[\frac{2\pi x\xi_0}{t} + \sqrt{4\pi^2\xi_0^2 + 1}\right]\right) \cdot \int_{\mathbb{R}} \exp\left(it \cdot \frac{2\pi^2}{(4\pi^2\xi_0^2 + 1)^{3/2}}(\xi - \xi_0)^2\right) d\xi$$

The latter integral, which we denote by J , evaluates to

$$J = \int_{\mathbb{R}} \exp\left(i \cdot \frac{2\pi^2(t^2 - x^2)^{3/2}}{t^2}(\xi - \xi_0)^2\right) d\xi = \left(\frac{2\pi}{|c|}\right)^{1/2} e^{i(\pi/4)},$$

where

$$c = \frac{4\pi^2(t^2 - x^2)^{3/2}}{t^2}$$

Therefore, one has that

$$|I_1| \approx |\hat{\phi}_0(\xi_0)| \cdot \left(\frac{t^2}{2\pi(t^2 - x^2)^{3/2}}\right)^{1/2} \sim \frac{1}{t^{1/2}}$$

where we use the fact that $\hat{\phi}_0$ is Schwartz, and thus, bounded. For $t > 1$, we have $1 + t \leq 2t \implies (1+t)^{1/2} \leq \sqrt{2}\sqrt{t} \implies \frac{\sqrt{2}}{(1+t)^{1/2}} \geq \frac{1}{t^{1/2}}$. Thus, $|I_1| \lesssim \frac{1}{(1+t)^{1/2}}$, where the constant was dependent only on the initial data. Note that we used the idea of *stationary phase approximations*¹.

¹https://en.wikipedia.org/wiki/Stationary_phase_approximation

(c) We can simply generalize the entire above argument to show that the desired decay rate is satisfied.

Problem E.1.9. Show that the bound $\sup_{t,x} |\partial\phi| \leq C$ for the solutions to the wave equation in $\mathbb{R} \times \mathbb{R}$ with smooth and compactly supported data is sharp in the following sense: suppose the initial data are given by $(\phi, \partial_t\phi) \upharpoonright_{\{t=0\}} = (\phi_0, \phi_1) \in C_c^\infty \times C_c^\infty$. Then if there exists a non-increasing function $A(t)$ such that $\lim_{t \rightarrow \infty} A(t) = 0$ and

$$(\sup_x |\partial\phi|)(t) \leq A(t),$$

then $\phi_0 = \phi_1 = 0$.

Let ϕ be a solution to the linear wave equation in $\mathbb{R} \times \mathbb{R}$ with compactly supported smooth data, and such that $\sup_{t,x} |\partial\phi| \leq C$. We recall d'Alembert's formula, which states that

$$\phi(t, x) = \frac{1}{2} [\phi_0(x+t) + \phi_0(x-t)] + \frac{1}{2} \int_{x-t}^{x+t} \phi_1(y) dy.$$

We then compute,

$$|\partial\phi|^2(t, x) = \frac{1}{2} [(\phi'_0)^2(x+t) + (\phi'_0)^2(x-t) + \phi_1^2(x+t) + \phi_1^2(x-t)] + \phi'_0(x+t)\phi_1(x+t) - \phi'_0(x-t)\phi_1(x-t).$$

Let $[a, b]^{\text{cpt}} \subseteq \mathbb{R}$ be a compact interval large enough to contain the support of the initial data. Fix an arbitrary point $x_0 \in [a, b]$, and let $x_1(t) = x_0 + t$ and $x_2(t) = x_0 - t$. Then it is easily shown that

$$\lim_{t \rightarrow \infty} |\partial\phi|^2(t, x_1(t)) = \frac{1}{2} [(\phi'_0)^2(x_0) + (\phi_1)^2(x_0) - \phi'_0(x_0)\phi_1(x_0)]$$

and

$$\lim_{t \rightarrow \infty} |\partial\phi|^2(t, x_2(t)) = \frac{1}{2} [(\phi'_0)^2(x_0) + (\phi_1)^2(x_0) + \phi'_0(x_0)\phi_1(x_0)].$$

By our hypothesis on the decay rate, we must also have $\lim_{t \rightarrow \infty} |\partial\phi|^2(t, x) = 0$ for every fixed $x \in \mathbb{R}$. Thus both of the above limits are zero. This yields $\phi'_0(x_0), \phi_1(x_0) = 0$. But since $x_0 \in [a, b]$ was arbitrary, $\phi'_0, \phi_1 \equiv 0$. Thus $\phi_0(x) \equiv c$ is a constant. But since the energy decays to zero as $t \rightarrow \infty$, but energy is also conserved, we must have $c \equiv 0$. Thus the conclusion follows.

Problem E.1.12. Show using the fundamental solution that solutions to the linear wave equation in $(1+1)$ -dimensions; i.e., $\mathbb{R} \times \mathbb{R}$, satisfy

$$\sup_{t,x} |\partial\phi| \leq C \sup_x |\partial\phi|(0, x)$$

for some $C > 0$ independent of ϕ . Show that the corresponding estimate fails in $\mathbb{R} \times \mathbb{R}^3$ (in fact, all dimensions $n \geq 2$!); i.e.,

$$\sup_{\{(\phi_0, \phi_1) : \sup_x |\partial\phi|(0, x) = 1\}} \left(\sup_{t,x} |\partial\phi| \right) = \infty.$$

This shows that unlike the L^2 norm for the derivatives of ϕ , which is propagated by the equation in all dimensions, the L^∞ norm for the derivative of ϕ is not propagated.

2.4 Random Computations

Problem 2. Analyze the modified energy, $E_p(t)$, defined by: for each $p \in (0, \infty)$, define

$$E_p(t) := \|\partial_t \phi\|_{L^p(\mathbb{R}^n)}^p(t) + \|\nabla \phi\|_{L^p(\mathbb{R}^n)}^p(t).$$

For $p = \infty$, define

$$E_\infty(t) = \|\partial_t \phi\|_{L^\infty(\mathbb{R}^n)}(t) + \|\nabla \phi\|_{L^\infty(\mathbb{R}^n)}(t).$$

Let ϕ be a sufficiently regular solution to the linear, homogeneous wave equation $\square \phi = 0$ on $\mathbb{R}^{1+n}$, and for $0 < p < \infty$, let

$$E_p(t) = \int_{\mathbb{R}^n} |\partial_t \phi|^p + |\nabla \phi|^p dx,$$

and for $p = \infty$, let

$$E_\infty(t) = \|\partial_t \phi\|_{L^\infty(\mathbb{R}^n)} + \|\nabla \phi\|_{L^\infty(\mathbb{R}^n)}.$$

We have already seen that when $p = 2$, $E_p(t)$ is constant in time for all $n \geq 1$. Our goal is to see if this conservation law extends to other p . First, let us consider $p = \infty$. We recall the pointwise dispersive estimates,

$$\sup_{x \in \mathbb{R}^n} |\phi(x, t)| \lesssim \frac{1}{(1+t)^{\frac{n-1}{2}}}.$$

Since derivatives commute, we have that $\partial_t \phi$ and $\nabla \phi$ are both solutions to the linear, homogeneous wave equation. So in fact, one has

$$\sup_{x \in \mathbb{R}^n} |\partial_t \phi(x, t)|, \sup_{x \in \mathbb{R}^n} |\nabla \phi(x, t)| \lesssim \frac{1}{(1+t)^{\frac{n-1}{2}}}.$$

Therefore for $n \geq 2$,

$$E_\infty(t) = \|\partial_t \phi\|_{L^\infty(\mathbb{R}^n)} + \|\nabla \phi\|_{L^\infty(\mathbb{R}^n)} \lesssim \frac{1}{(1+t)^{\frac{n-1}{2}}} \rightarrow 0 \text{ as } t \rightarrow \infty,$$

so that $E_\infty(t)$ is not constant in time. Now consider $n = 1$. We can easily disprove the conservation law via an example. Consider the Cauchy wave problem,

$$\begin{cases} \square \phi = 0, \\ (\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = ((-1+x^2)\chi_{[-1,1]}, 0) \end{cases}$$

Then $E_\infty(0) = \|\nabla \phi\|_{L^\infty}(0) = 2$. However, in the period $0 < t < 1/2$, $\|\nabla \phi\|_{L^\infty(\mathbb{R})} > 1$ so that $E_\infty(t) > 2$ for all $t \in (0, 1/2)$. Thus $E_\infty(t)$ cannot be conserved. This rules out any energy conservation law for $p = \infty$.

Now let us consider $p = 1$,

$$E_1(t) = \int_{\mathbb{R}^n} |\partial_t \phi| + |\nabla \phi| dx = \|\partial_t \phi\|_{L^1(\mathbb{R})} + \|\nabla \phi\|_{L^1(\mathbb{R})}.$$

We will consider the case $n = 1$ first. Consider the Cauchy wave problem,

$$\begin{cases} \square \phi = 0, \\ (\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = ((-1+|x|)\chi_{[-1,1]}, 0) \end{cases}$$

Then $E_1(0) = \|\nabla \phi\|_{L^1(\mathbb{R})} = 2$. Now let $t \in (0, 1/2)$. Then

$$\begin{aligned} \|\nabla \phi\|_{L^1(\mathbb{R})}(t) &= \int_{-1-t}^{-1+t} \frac{1}{2} dt + \int_{-1+t}^{-t} dt + \int_t^{1-t} dt + \int_{1-t}^{1+t} \frac{1}{2} dt \\ &= 2 - 2t. \end{aligned}$$

Likewise, we compute $\|\partial_t \phi\|_{L^1(\mathbb{R})}(t) = 4t$. Therefore, $E_1(t) = 4t + 2 - 2t = 2 + 2t > 2$. Evidently, $E_1(t)$ is not, in general, conserved in $n = 1$. Now we consider $n \geq 2$. We modify the Cauchy problem from above as follows,

$$\begin{cases} \square \phi = 0, \\ (\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = ((-1 + |x|)\chi_{\{|x| \leq 1\}}, 0), \end{cases}$$

where $x = (x_1, \dots, x_n) \in \mathbb{R}^n$. Then we compute $E_1(0) = \|\nabla \phi\|_{L^1(\mathbb{R}^n)}(0) = \omega_n$, where ω_n is the volume of the unit n -ball. On the other hand, for $t \in (0, 1/2)$, switching to radial coordinates and using the dispersion of solutions to the wave equation, we can show that the energy exceeds the initial energy. Thus $E_1(t)$ is not conserved for any $n \geq 2$. This rules out any energy conservation law for $p = 1$. Going one step further, we have the following.

Definition 2.1 (Energy). *Let $n \geq 1$ be a positive integer, and let ϕ be a solution to the linear, homogeneous wave equation*

$$\begin{cases} \square \phi = 0, \\ (\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = ((-1 + |x|)\chi_{\{|x| \leq 1\}}, 0) \end{cases} \quad (1)$$

in $\mathbb{R}^{1+n}$. For each $p \in (0, \infty)$, define

$$E_p(t) := \|\partial_t \phi\|_{L^p(\mathbb{R}^n)}^p(t) + \|\nabla \phi\|_{L^p(\mathbb{R}^n)}^p(t).$$

For $p = \infty$, define

$$E_\infty(t) = \|\partial_t \phi\|_{L^\infty(\mathbb{R}^n)}(t) + \|\nabla \phi\|_{L^\infty(\mathbb{R}^n)}(t).$$

Conjecture. *Let $n \geq 1$ be a positive integer, and let ϕ be a (nontrivial) solution to (1). For all $p \in (0, \infty] \setminus \{2\}$,*

$$\sup_{t \in (0, 1/2)} E_p(t) > E_p(0).$$

If $p = 2$, then $E_p(t)$ is constant for all $t \geq 0$.